

Number system - Hexa-decimal number (Base-16)

Any numerical value (number) can be represented in any of the following number system.

Number Base	Number system	
Base 2	binary	The common representation of data in computer is a string of bits, using the binary numeral system (Base-2).
Base 8	octal	
Base 10	decimal	The most familiar representation of numbers is in base 10 systems. Eg 12345.
Base 16	hexadecimal	Binary strings (01000001) are long and hard for human programmers to read. To simplify this, programmers split a byte in half (two 4-bit sections called nibbles) and convert them into Hexadecimal (Base-16). It is convenient shorthand notation to talk about binary numbers. Hexadecimal is also commonly used to represent computer memory addresses.

Hexadecimal is a number system that uses 16 symbols to represent numbers.

- Hexadecimal uses the symbols 0–9 to represent the values 0–9.
- It uses the letters A–F to represent the values 10–15. In hexadecimal, when you reach the decimal number 10, you switch to letters A through F because a single digit can hold any value up to 15.
- Each digit represents a decimal value. For example, D is equal to 13 in base 10.

Decimal (Base 10)	Hexadecimal (Base 16)	4-Bit Binary Value
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

Conversion between number systems

<u>Decimal to Hexa-decimal conversion</u>									<u>Hex to decimal conversion</u>					
Identify the group of 4 bits each. Fill the partial group with 0's on outside. Then replace each group with the corresponding hex digit.									hexadecimal number (Base-16) 0x2AF3 is equal to 10,995 decimals (base-10).					
0011	1010	0101	0110	.	1110	0010	1111	1000	Base-16	2	A	F	3	
3	A	5	6	.	E	2	F	8		base-16 to base-10	2×16^3	$A \times 16^2$	$F \times 16^1$	3×16^0
											2×16^3	10×16^2	15×16^1	3×16^0
											2x4096 =8192	10x256 =2560	15x16 =240	3*1 =3
									Base-10	10,995				

8-Bits (1byte)	1	1	1	1	1	1	1	1
Base -2 place holder	$1 \times 2^7 =$ (128)	$1 \times 2^6 =$ (64)	$1 \times 2^5 =$ (32)	$1 \times 2^4 =$ (16)	$1 \times 2^3 =$ (8)	$1 \times 2^2 =$ (4)	$1 \times 2^1 =$ (2)	$1 \times 2^0 =$ (1)
Decimal value	255							
Hexadecimal value	0xFF							